# Brain Stroke Detection Using Deep Transfer Learning

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***Abstract*— One of the most serious medical emergencies and a major global cause of death and permanent disability is brain stroke. Reducing neurological damage and increasing survival rates require prompt and precise diagnosis. Because of their speed and accessibility, computed tomography (CT) scans are frequently used to diagnose strokes. However, interpretation calls for skilled radiologists, who may not be available in rural and semi-urban healthcare facilities. An AI-assisted brain stroke detection system that applies deep transfer learning techniques to CT scan images is presented in this paper. For binary stroke detection, severity grading, and explainability using Gradient-weighted Class Activation Mapping (Grad-CAM), the suggested system uses MobileNetV2, a lightweight convolutional neural network. To improve diagnostic reliability, clinical parameters like age, body mass index, diabetes, hypertension, and smoking history are combined. In order to facilitate real-time, AI-assisted triage, the system is implemented as a cloud-based web application. The suggested method enhances early diagnosis, interpretability, and practical usability in clinical settings with limited resources, according to experimental analysis.***

***Keywords: Grad-CAM, CT Scan, Deep Transfer Learning, Brain Stroke Detection, Severity Grading, and Medical Imaging.***

## I. Introduction

The need for automated diagnostic systems to help physicians make timely decisions has grown dramatically as a result of the exponential growth of medical imaging data. Brain stroke happens when a portion of the brain's blood supply is cut off or reduced, depriving the brain's tissue of nutrients and oxygen. Stroke can cause death or irreversible brain damage if it is not diagnosed and treated quickly [1]. Improving patient outcomes requires early detection during the "golden hour" [2].

Because of their quick acquisition and capacity to distinguish between ischemic and hemorrhagic strokes, computed tomography (CT) scans are the first-line imaging modality for stroke diagnosis [3]. However, qualified radiologists are needed for accurate CT scan interpretation, and they are scarce in rural and semi-urban healthcare facilities [4]. Increased mortality rates and delayed diagnosis are frequent outcomes of this gap.

Deep learning and artificial intelligence have shown great promise in medical image analysis, allowing for the automated and highly accurate detection of abnormalities [5]. Specifically, transfer learning enables the adaptation of pre-trained models for medical imaging tasks with small labeled datasets [6]. Despite these developments, a large number of stroke detection systems still in use only use binary classification and lack explainability, real-time deployment, and severity assessment [7].

Furthermore, in addition to imaging, clinical decision-making for stroke diagnosis also takes into account patient-specific risk factors like age, diabetes, hypertension, and smoking [8]. Systems that disregard these elements might not accurately represent clinical situations in the real world. Furthermore, adoption is hampered and clinician trust is diminished by black-box AI models that lack interpretability [9]. These difficulties drive the creation of a multimodal, deployable, clinically explicable AI-based stroke detection system [10].

### 1.1 Context and Challenges:

In modern healthcare systems, there is a need to balance the speed and accuracy of diagnosis with the availability of resources. In the case of strokes, there is a need to ensure that diagnosis occurs within a reasonable time to improve outcomes. The main challenges include a lack of access to skilled radiologists, inconsistent quality of CT scans, and the complexity of interpreting subtle patterns associated with strokes. The deep learning model needs to generalize well across different populations and conditions while being computationally efficient.

The second major problem associated with deep learning models in medical diagnosis is the lack of transparency in these models. There is a need to ensure that clinicians can understand the predictions made by AI systems to improve their reliability and confidence in these predictions. Moreover, most automated systems do not consider clinical factors, which play a critical role.

### 1.2 Data Overview:

The proposed system uses images obtained from brain CT scans. The images have been obtained from publicly available datasets. The dataset includes images with stroke as well as images without stroke. The images with stroke include images of hemorrhagic as well as ischemic stroke. In addition to images, clinical data such as patient age, blood glucose levels, hypertension, diabetes, body mass index, and smoking habits have also been incorporated.

## II. LITERATURE SURVEY

### 2.1 Early Rule-Baed Medical Imaging Systems

Early Early automated medical image processing systems were mostly rule-based and used classical image processing techniques like thresholding, edge detection, region growing, and morphological transformations. The rule-based systems used expertise in the medical field to develop explicit rules to recognize abnormalities in medical images. In brain image processing, intensity thresholds were used to separate healthy and abnormal regions in images, and shape-based rules were used to locate abnormal regions.

Although rule-based image processing systems provided advantages like interpretability and lower computational complexity, these systems were highly sensitive to image quality and conditions. The accuracy of these systems decreased with changes in CT scan settings, differences in human body shapes, and levels of illumination.

Moreover, these systems required a lot of tuning for different images and image acquisition conditions, making them difficult to apply universally. Hence, rule-based image processing systems were found to be inaccurate and unreliable for detecting strokes, making it difficult to apply them in practical scenarios [11].

### 2.2 Machine Learning Approaches for Stroke Detection

With To circumvent the problems associated with rule-based systems, traditional machine learning techniques were incorporated for stroke detection. The traditional machine learning techniques employed feature extraction techniques such as texture features, intensity histogram features, wavelet features, and statistical features. The extracted features were further used to train classifiers such as SVM, KNN, Decision Trees, and Random Forests.

The traditional machine learning techniques showed better accuracy and flexibility in comparison to rule-based techniques. Nevertheless, the accuracy of traditional machine learning techniques was largely dependent on the quality and relevance of manually extracted features. The quality of features was highly dependent on domain expertise, as creating good feature sets was not easy. In addition, these models did not generalize well when applied to large and varied datasets, since the features were not robust to changes in stroke appearance, size, and location. These limitations paved the way for automated feature learning approaches, which led to the development of stroke detection systems based on deep learning techniques [12].

### 2.3 Transfer Learning for CT Scan Analysis

The advent of CNNs has significantly impacted medical image analysis, allowing for the automation of image feature learning. The adoption of architectures like VGG, ResNet, and Inception for brain abnormality detection, including stroke classification, has shown promising results [13].

### 2.4 Explainable AI in Healthcare

The importance of Explainable Artificial Intelligence (XAI) has become a critical requirement for healthcare applications, where model interpretability and trustworthiness are as important as model accuracy. Deep learning models, especially convolutional neural networks, are often accused of being “black boxes,” which make it difficult for clinicians to understand the basis of their predictions. The lack of interpretability of deep learning models has become a major impediment to their adoption, especially in high-risk applications such as stroke diagnosis. Methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) have provided a solution to this problem by generating visual maps that highlight parts of an input image that contribute to the final decision of a model. The method has been successfully applied to medical image analysis tasks such as pneumonia detection, brain tumor segmentation, and diabetic retinopathy analysis.

In this way, explainable AI techniques have a vital role to play in bridging the gap between deep learning techniques and reality in clinical practice [14][15].

### 2.5 Stroke Severity Assessment

While significant advancements have been made in the automated detection of stroke, most of these computer-assisted diagnosis techniques have only been capable of performing binary classification, i.e., distinguishing between stroke and non-stroke. However, in reality, the assessment of stroke severity is equally important, as this directly impacts the triage, treatment, and management of stroke. The severity grading helps in determining the urgency of the situation as well as the distribution of resources.

Automated stroke severity assessment is an underexplored field with many challenges, including the varying sizes, positions, and intensities of lesions among patients. Current methods involve the use of clinical scales such as the National Institutes of Health Stroke Scale (NIHSS).

The addition of severity grading to AI-based systems may prove to be beneficial from a clinical perspective, enabling early risk assessment prior to expert analysis. Studies suggest the addition of lesion characteristics and activation from deep learning models may prove useful in the estimation of the the severity of the problem[16][17].

.2.7 Deployment Challenges in Clinical AI Systems

The practical application of artificial intelligence models in real-world healthcare settings also raises a number of issues that go beyond model accuracy. For example, there is the concern of inference latency in healthcare settings. This is because clinical decisions require prompt responses, especially in emergency situations such as stroke detection. As a result, AI models have to provide responses within very short time constraints in these situations. In addition, there is the concern of data privacy. This is because medical information is very sensitive and has to comply with very strict regulations. Furthermore, scalability is also a concern in the practical application of AI models in healthcare settings. This is because these models can be accessed by many healthcare centers at the same time. However, cloud computing has been identified as a possible solution to these challenges. This is because cloud computing can provide a platform for hosting AI models in healthcare settings without the need for expensive hardware[18].

2.8 Reliability and Generalization Issues

Analyses Even though the models perform well on benchmark sets, they do not generalize well to other environments. This is because of the different manufacturers of CT scanners, which can affect the performance of the models. Moreover, there is a possibility of dataset bias, which can occur due to insufficient training. This can, therefore, make the models less reliable. To address these problems, strong preprocessing methods can be used. This includes normalization, augmentation, and the use of a constant resolution. Splitting, cross-validation, and validation of the models can also be used. All these problems need to be addressed to ensure that the AI-based stroke detection systems are reliable and can be used anywhere [19].

Summary of Literature

The Moreover, the literature review suggests that the progression from conventional rule-based approaches and traditional machine learning techniques to deep learning and transfer learning-based approaches is evident for medical image analysis. Although the state-of-the-art deep learning-based approaches have reported high accuracy for brain stroke detection tasks, some limitations still need to be addressed.

The existing studies have reported the importance of lightweight models, explainability, and the integration of clinical data into the brain stroke detection systems. These observations from the literature review collectively form the basis for the development of the proposed brain stroke detection system that addresses the limitations of the existing approaches by being explainable, multimodal, and deployable for brain stroke detection [20].

In addition to this, the literature also shows that the majority of the studies conducted on the detection of strokes have remained limited to the experimental environment and have not been implemented as a system.

## III. Proposed Model

The proposed model offers a comprehensive and flexible model for automated brain stroke detection through deep transfer learning. The model incorporates image analysis from a CT scan and clinical risk assessment, along with explainable AI, to aid in the early diagnosis of brain strokes. The model's architecture allows it to be scalable and lightweight, making it suitable for implementation in low-resource medical settings such as rural and semi-urban medical centers. The model utilizes a lightweight CNN and cloud-based inference to reduce delays in diagnosis while being transparent and trustworthy.

3.1 System Framework

The system utilizes a client-server model that consists of three major components: a web-based user interface, a backend AI processing component, and a cloud-based inference engine. The user interface component offers a simple and intuitive user experience that can be used to upload brain CT scan images and other relevant clinical information by healthcare personnel.

The backend component will be used to process the uploaded information and results aggregation. The pre-trained MobileNetV2 model will be used as the primary component for deep learning image analysis.

Other components will be added to process clinical information and model interpretability. The system will be implemented using a cloud-based model to host the inference engine and other components, reducing the need to purchase expensive hardware to host the model locally. The components will be implemented in a manner that can be accessed remotely and can be easily scaled to accommodate a larger number of users.

The backend component will be implemented to process the uploaded images and results aggregation. The pre-trained MobileNetV2 model will be used as the primary component for deep learning image analysis. Other components will be added to process clinical information and model interpretability.

3.2 Operational Workflow

The operational workflow of the system starts from the point when a user uploads a CT scan using the web interface. The uploaded CT scan is then taken through a series of preprocessing steps before it is fed into the deep learning model for the purpose of binary classification. In the binary classification process, the MobileNetV2-based classifier is used to identify the presence or absence of a stroke in the CT scan uploaded by the user. For stroke-positive patients, the system is able to carry out the grading of the severity of the stroke. In addition to that, the Grad-CAM module is used to generate a visual explanation of the decision made by the deep learning model.

At the same time, the clinical data uploaded by the user is taken through a process to calculate the risk score for the patient.

The system processes user-uploaded CT scans through preprocessing, deep transfer learning-based stroke detection, severity grading, clinical data fusion, and Grad-CAM visualization to produce a real-time explainable output.

### 3.3 Decision Fusion Mechanism

To enhance the reliability of the diagnosis, a decision fusion technique is used to combine the predictions made by the imaging model and clinical model. The imaging model will be used to detect abnormalities in the CT scan image, while the clinical model will be used to assess the clinical risk factors, which include age, hypertension, diabetes, body mass index, and smoking.

The final diagnosis will be made by fusing the predictions made by the imaging model and clinical model, and this will be achieved by applying a weighted decision fusion technique. This will help to reduce false positives and false negatives by matching the predictions made by the model with the actual human reasoning process.

The model will be able to take a more holistic view by considering both the visual and contextual information provided

### 3.4 Future Enhancements

improvements to the proposed model involve the integration of the model into the hospital's information systems for efficient data exchange and record-keeping purposes. In addition, the integration of other forms of medical images, such as MRI scans, into the model could lead to better diagnostic outcomes and severity levels.

The integration of the automated severity grading into the NIH Stroke Scale is another area for validation. Optimizations of the user interface and security mechanisms will also be considered for the model.

## IV. Methodology

### 1.1 Deployment and Implementation:

_The brain stroke detection system, as proposed, can be implemented with a backend system based on the Flask framework, which can act as the central processing unit of the system.

The backend system can be implemented with the Flask framework because of its lightweight architecture, flexibility, and ability to support various machine learning and deep learning frameworks, including TensorFlow and PyTorch, which can be used for brain stroke detection.

The MobileNetV2 model, which has already been pre-trained on the ImageNet dataset, can be fine-tuned with stroke-related CT scan images for binary classification of brain stroke and non-stroke cases.

The top layers of the MobileNetV2 model can be replaced with fully connected layers, which can be used to fine-tune the model with brain stroke-related CT scan images. This can help reduce the training time of the model, as well as provide high accuracy for brain stroke classification.

The Grad-CAM is incorporated into the inference pipeline to improve model interpretability. The Grad-CAM creates heat maps during the prediction that provide a visual representation of the parts of the image contributing to the classification. The heat maps are overlaid on the original images of the CT scans and provided to clinicians for validation of the predictions.

The clinical parameters such as age, blood glucose levels, hypertension, diabetes, body mass index, and smoking history are processed using structured machine learning to calculate a patient-specific risk score. The proposed system is deployed on a cloud platform to provide real-time access while reducing infrastructure requirements at healthcare centers.

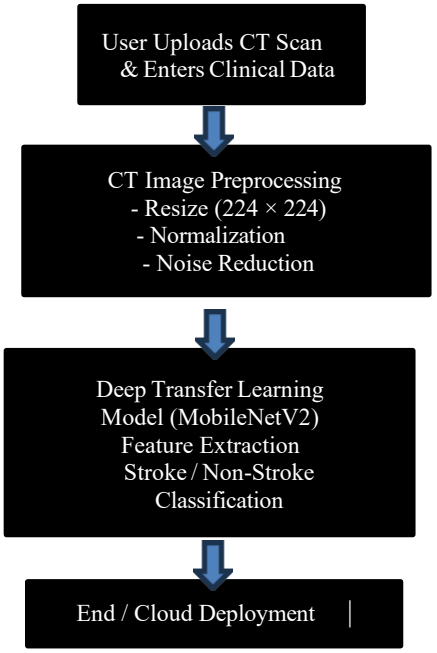



Fig.1 : Methodology workflow

## V. Result and Discussion

The proposed brain stroke detection system was evaluated to check the classification accuracy, interpretability, and computational efficiency of the system. The experimental results of the brain stroke detection system indicate that the accuracy of the system, based on the deep transfer learning model, is high for differentiating between stroke and non-stroke CT scan images.

performance of the brain stroke detection system was evaluated by using various classification performance metrics, including accuracy, precision, recall, and F1-score. The results of the classification performance metrics indicate that the brain stroke detection system can accurately classify brain stroke cases with low false negatives. This is because the brain stroke detection system can correctly classify brain stroke cases, as confirmed by the results of the confusion matrix. In addition to binary classification, the system's stroke severity grading module offers further insight by grading the detected strokes as minor, moderate, and severe. This helps in the efficient management of the detected strokes. Though the system's estimation of the severity of the stroke is approximate, it serves as preliminary data before the expert's final evaluation. In the proposed system, the results of the interpretability using the Grad-CAM algorithm play a vital role in validating the results and other related data provided by the model

Table 1 : STROKE DETECTION SYSTEM MODULES

| Module | Function Overview | Input | Output |
|---|---|---|---|
| Image Acquisition | Captures CT scan slices for analysis. | DICOM/ PNG CT images | Raw CT frames |
| Preprocessing | Resizes, normalizes, and augments CT images. | Raw CT frames | Preprocessed images |
| Feature Extraction (MobileNetV2) | Extracts deep feature representations from CT scans. | Preprocessed image | Feature vectors |
| Classification Head | Performs stroke vs. non-stroke classification and severity grading. | Feature vectors | Stroke label + severity |
| Clinical Risk Model | Computes risk score from clinical parameters (age, BMI, etc.). | Tabular clinical data | Risk score |
| Decision Fusion Layer | Combines imaging predictions and clinical risk score. | Stroke/severity + risk score | Final diagnosis / triage suggestion |

## V. Conclusion

This study proposed an AI-based framework for the detection of brain stroke using deep transfer learning, which could be useful in the early diagnosis and decision-making processes in real-world healthcare settings. The framework effectively differentiates between stroke and non-stroke conditions using the imaging results from the CT scan and the lightweight CNN model, without compromising the speed and computational cost of the system. The use of MobileNetV2 in the proposed framework makes it feasible to deploy the system in real-world settings, such as rural and semi-urban hospitals, where expert radiologists may not be readily available.

The novelty in the proposed framework is the use of the concept of explainable AI with the implementation of the Grad-CAM technique, which provides visual interpretations of the AI model's predictions and decisions. This visual explanation helps clinicians validate the results and verify whether the AI model is focusing on the relevant parts of the brain during the diagnosis process. The proposed framework also includes the functionality to determine the severity level and risk factors related to the occurrence of the stroke, providing deeper and more relevant information regarding the diagnosis process.

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